Year 7 Mathematics

Unit 2: Number Skills

2.4 Squares and Square Roots

Squares and Square Roots

STARTER · 5 MIN

Date: _____ **Title: Squares and Square Roots**

LEARNING OBJECTIVES

- To calculate with squares and square roots and use index notation for it

STARTER *Answer in your book. No calculators.*

1. Work out 4×4
2. Work out 7×7
3. What number times itself is 81?
4. Work out 10×10

Starter — Answers

STARTER

Tick or correct in green pen

1. **16**

2. **49**

3. **9**

4. **100**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can identify and use square and square-root notation	☐
SILVER	I can calculate squares and square roots	☐
GOLD	I can apply squares and square roots to solve problems	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Squares and Square Roots

INTRODUCTION · 3 MIN

A square number is a number multiplied by itself.

Square number

A number times itself.

$4^2 = 4 \times 4 = 16$

Index notation

The small 2 is the power (index).

5^2 is '5 squared'

Square root

The inverse of squaring.

$\sqrt{16} = 4$ because $4^2 = 16$

Area

Area of a square = side^2.

Side 6 cm: area = 36 cm^2

Examples 1 and 2 — Squaring Numbers

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint $6^2 = 6 \times 6$ Squared: times itself

Example 1

Work out

(a) 6^2

(b) 11^2

Example 2

Write 7×7 using index notation. Work it out.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Square: multiply by itself

(a) $6^2 = 6 \times 6$

$= 36$

(b) $11^2 = 11 \times 11$

$\mathbf{= 121}$

Tip: 6^2 is not 6×2.

Example 2

7 × 7 is 7 squared

$7 \times 7 = 7^2$

$\mathbf{= 49}$

Tip: the small 2 is the index.

Task 1

PRACTICE · 9 MIN

BRONZE *Work out each square*

Hint $6^2 = 6 \times 6$ Squared: times itself

1. 3^2

2. 8^2

3. 12^2

4. Write 9×9 in index notation

5. 15^2

★ $1^2 + 2^2 + 3^2$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. 9
2. 64
3. 144
4. $9^2 = 81$
5. 225

★ 1 + 4 + 9 = 14

Examples 3 and 4 — Square Roots

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint Square root: undo squaring $7^2 = 49$, so the root of 49 is 7

Example 3

Work out

(a) $\sqrt{49}$

(b) $\sqrt{144}$

Example 4

Between which two whole numbers is the square root of 50?

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Which number squared gives it?

(a) $\sqrt{49} = 7$

(b) $\sqrt{144} = 12$

Tip: $7^2 = 49$ and $12^2 = 144$.

Example 4

Find the squares either side

$7^2 = 49$ and $8^2 = 64$

50 is between 49 and 64,

so the root is between 7 and 8.

Tip: 50 is close to 49, so about 7.1.

Task 2

PRACTICE · 9 MIN

SILVER *Work out each square root*

Hint Square root: undo squaring $7^2 = 49$, so the root of 49 is 7

1 $\sqrt{25}$

2 $\sqrt{81}$

3 $\sqrt{121}$

4 $\sqrt{1}$

5 Between which two whole numbers is the root of 200?

★ $\sqrt{9 \times 16}$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. **5**

2. **9**

3. **11**

4. **1**

5. **14 and 15**

★ $\sqrt{144} = 12$

Examples 5 and 6 — Squares and Roots in Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint Side = root of the area Powers and roots first

Example 5

(2 marks)

A square has an area of 196 cm^2.

Work out its side length.

Example 6

(2 marks)

Work out

$5^2 + \sqrt{36}$

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Side = square root of the area

Side $= \sqrt{196}$

$= 14$ cm

Tip: check 14 × 14 = 196.

Example 6

Powers and roots first

$5^2 = 25$

$\sqrt{36} = 6$

Total $= 25 + 6$

$= 31$

Tip: work out each part, then add.

Task 3

PRACTICE · 9 MIN

GOLD *Exam-style: show all your working*

Hint Side = root of the area Powers and roots first

1 A square has area 64 cm^2.
Find its side length.

2 A square has side 13 cm.
Find its area.

3 2×6^2

4 $\sqrt{100} - 3^2$

5 Find two square numbers that add up to 25.

★ A square field has area 400 m^2.
Find its perimeter.

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. 8 cm

2. 169 cm²

3. 2 × 36 = 72

4. 10 − 9 = 1

5. 9 + 16 = 25

★ Side = 20 m

Perimeter = 4 × 20 = 80 m

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Work out 9^2

SILVER Work out $\sqrt{49}$

GOLD A square has area 225 cm^2. Find its side length.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **81**

SILVER **7**

GOLD **15 cm**

Exam Practice — Squares

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint $6^2 = 6 \times 6$ Squared: times itself

Question 1

(2 marks)

Work out each square.

(a) 9^2 (1)

.....

(b) 13^2 (1)

.....

Question 2

(2 marks)

Index notation.

(a) Write 6×6 using index notation. (1)

.....

(b) Work it out. (1)

.....

Exam Practice — Square Roots

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint Square root: undo squaring
$7^2 = 49$, so the root of 49 is 7

Question 3

(2 marks)

Work out each square root.

(a) $\sqrt{144}$ (1)

.....

(b) $\sqrt{64}$ (1)

.....

Question 4

(2 marks)

Estimating square roots.

(a) Between which whole numbers is $\sqrt{30}$? (1)

.....

(b) Which is bigger: $\sqrt{50}$ or 7? (1)

.....

Exam Practice — Problems

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint Side = root of the area
Powers and roots first

Question 5

(3 marks)

A square has an area of 196 cm^2.

(a) Work out its side length. (2)

..... cm

(b) Work out its perimeter. (1)

..... cm

Question 6

(2 marks)

Work out

$4^2 + \sqrt{25} \times 2$

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 81 B1

(b) 169 B1

Tip: 13 × 13 = 130 + 39.

Question 2

(a) 6^2 B1

(b) 36 B1

Tip: the index 2 means squared.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) 12 B1

(b) 8 B1

Tip: $12^2 = 144$.

Question 4

(a) 5 and 6 B1

(b) $\sqrt{50}$, as $7^2 = 49$ B1

Tip: $5^2 = 25$ and $6^2 = 36$.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) Side $= \sqrt{196}$ — M1

$= 14$ cm — A1

(b) Perimeter $= 4 \times 14 = 56$ cm — B1

Tip: perimeter = 4 sides.

Question 6

$16 + 5 \times 2 = 16 + 10$ — M1

$= 26$ — A1

Tip: × before +.